Kalvakuntla Ritanshu Kumar

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Education

Vellore Institute of Technology (VIT) — Vellore, India

Integrated M.Tech in Computer Science and Engineering – Data Science ***(CGPA: 8.06)*** — Aug 2022 – Sept 2027

Experience

Machine Learning Intern — June 2025 – July 2025

MANMUN PVT LTD, Hyderabad

- Worked on AI-driven application development with practical exposure to Machine Learning and Deep Learning model implementation, testing, and evaluation.
- Developed and evaluated multiple machine learning models, gaining hands-on experience across the end-to-end model development workflow.

FormWise AI | GitHub | Demo

VIBE Coding Competition → Independent Product Development

- Built an AI-powered form builder and feedback analytics platform during the VIBE Coding Competition, then independently extended it into a fully deployed production application.
- Integrated Google Gemini for response summaries, sentiment, themes, and insights; built REST APIs with FastAPI and PostgreSQL and a Next.js/TypeScript frontend, deployed using Vercel, Render, and Supabase.

Projects

Document Q&A RAG Assistant | GitHub | *Python, ChromaDB, Llama 3.1/Groq API, SQLite, Streamlit*

- Built a retrieval-augmented document QA system using ChromaDB vector search and Llama 3.1 via Groq API for grounded responses.
- Implemented document ingestion, chunking, embedding retrieval, prompt construction, and SQLite-based evaluation logging.

Hybrid CNN-Transformer Model for Pneumonia Detection | GitHub | *Python, PyTorch, Flask, Grad-CAM*

- Built HybridXRayNet combining multi-scale CNN blocks, residual attention, Transformer Encoder, adaptive feature fusion, and Grad-CAM for interpretable pneumonia detection.
- Achieved 83.17% accuracy, 79.26% precision, 98.97% recall, 88.03% F1-score, and 96.67% ROC-AUC on a 624-image held-out test set.

CAPTCHA Recognition using CRNN + CTC | GitHub | *Python, PyTorch, ResNet18, BiGRU, CTC*

- Developed an end-to-end CAPTCHA recognition system combining ResNet18 visual feature extraction, bidirectional GRU sequence modeling, and CTC decoding without explicit character segmentation.
- Trained and evaluated on 9,955 CAPTCHA images, achieving **99.30% exact-string accuracy** and **99.86% character-level accuracy** on a 1,991-image held-out test set.

Open Source Contributions

Expense Tracker | *HTML, CSS, JavaScript, Chart.js* | GitHub PR

- Implemented a dark mode toggle using CSS custom properties and localStorage-based theme persistence.
- Resolved a merge conflict and collaborated with the maintainer through code review; contribution was reviewed and merged.

Technical Skills

Programming & Data Science: Python, SQL, C++, JavaScript, Pandas, NumPy, Scikit-learn, EDA, Statistics, Feature Selection, Data Visualization, Power BI

AI/ML & Deep Learning: PyTorch, TensorFlow/Keras, CNNs, Transformers, OpenCV, NLP, Computer Vision, LLMs, RAG, Embeddings, Vector Databases, Prompt Engineering

Web Development & APIs: React.js, FastAPI, Flask, Streamlit, Node.js, Express.js, MongoDB, PostgreSQL

Tools & DevOps: Git, GitHub, Docker, Apache Airflow, AWS (EC2, S3), Vercel

Certifications

- **100x Dev**: Live Full Stack Open Source Cohort-1
- **Coursera**: Supervised Machine Learning
- **Forage**: Tata GenAI Powered Data Analytics Job Simulation
- **Embrizon**: Machine Learning with Python